Loan-to-Value Methodology for Margin Trading

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Abstract

This document defines the quantitative methodology governing the calculation, monitoring, and enforcement of Loan-to-Value (LTV) ratios within the margin trading product. It serves as the contractual and operational reference between the Platform and the external Lender, covering collateral eligibility, haircut derivation, LTV computation, threshold calibration, dynamic regime adjustment, margin call procedures, forced liquidation logic, stress testing, and governance. The framework is designed to be mathematically rigorous, operationally transparent, and aligned with applicable regulatory standards (MiFID II, CRR II, ESMA leverage guidelines).

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Contents

1 Purpose and Scope

1.1 Purpose

This methodology document establishes the quantitative framework governing Loan-to-Value (LTV) ratios for the margin trading product offered by the Platform. It provides a single authoritative reference for:

- the mathematical definition and computation of LTV;
- the derivation of collateral haircuts using risk-based methods;
- dynamic threshold adjustment under changing market regimes;
- margin call and forced liquidation procedures;
- stress testing and scenario analysis of the collateral portfolio;
- governance, backtesting, and model validation requirements.

1.2 Scope

This methodology applies to all margin loans extended to retail and professional clients, all collateral held in segregated custody accounts, and all lifecycle events affecting the loan balance or collateral portfolio. It does *not* cover unsecured lending, short-selling margin requirements, or derivatives margining.

1.3 Regulatory Context

The framework is aligned with:

- **MiFID II / MiFIR** – suitability and leverage requirements;
- **CRR II / Basel III** – Financial Collateral Comprehensive Method (FCCM) for haircut recognition;
- **ESMA Guidelines on Leverage** – retail client leverage caps by asset class;
- **EMIR** – where collateral overlaps with derivative positions.

2 Definitions

Definition 2.1 (Loan Amount)**.** *The loan amount at time t is denoted L_t and comprises outstanding principal plus accrued interest.*

Definition 2.2 (Gross Market Value)**.** *The gross market value of the collateral portfolio at time t is*

$$V_t^{\text{gross}} = \sum_{i=1}^{N} P_{i,t}\, Q_i,$$

where $P_{i,t}$ is the last traded price of asset i and Q_i is the quantity held.

Definition 2.3 (Adjusted Collateral Value). *The adjusted collateral value V_t^{adj} is the gross market value after application of per-asset haircuts, concentration caps, and a portfolio-level correlation adjustment (defined in Section* **??***).*

Definition 2.4 (Loan-to-Value Ratio).

$$\text{LTV}_t = \frac{L_t}{V_t^{\text{adj}}}$$

Definition 2.5 (Haircut). *The haircut $h_i \in [0, 1)$ for asset i is the proportional reduction applied to its market value to derive its eligible collateral contribution, reflecting liquidation risk.*

Definition 2.6 (Concentration Adjustment). *A portfolio-level multiplicative factor $\kappa_t \in [0, 1)$ applied to V_t^{adj} to penalise undiversified or highly correlated collateral portfolios.*

3 Collateral Eligibility and Haircut Framework

3.1 Eligibility Criteria

An asset qualifies as eligible collateral if and only if it satisfies *all* of the following criteria simultaneously:

1. **Listing:** Traded on a recognised exchange (EU regulated market or equivalent third-country venue).
2. **Liquidity:** 30-day average daily traded value (ADTV) $\geq$ 500,000.
3. **Price Availability:** End-of-day and intraday prices available from a primary source and at least one independent fallback.
4. **Instrument Type:** Equities, broad-market ETFs, government bonds, investment-grade corporate bonds. Explicitly excluded: structured products, leveraged/inverse ETFs, unlisted securities, crypto-assets.
5. **Jurisdiction:** Issuer domiciled or listed in EEA, US, UK, Switzerland, Japan, Australia, or Canada.

3.2 Risk-Based Haircut Derivation

Haircuts are designed to absorb the potential mark-to-market loss over the liquidation horizon with high probability. The base haircut for asset i is derived as the Value-at-Risk of a long position over a holding period Δt:

$$h_i = 1 - \frac{1}{1 + r_f\,\Delta t}\,\exp\!\Big(-\,z_\alpha\,\hat{\sigma}_i\,\sqrt{\Delta t}\Big)\,, \tag{1}$$

where:

- $\hat{\sigma}_i$ – annualised return volatility of asset i, estimated via an EWMA model with decay factor $\lambda = 0.94$ over a 252-day window:

  $$\hat{\sigma}_i^2 = (1-\lambda)\sum_{k=0}^{T-1}\lambda^k r_{i,t-k}^2,$$

  with daily log-returns $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$;

- $\Delta t = 5/252$ – assumed liquidation horizon of five trading days;
- $z_\alpha = \Phi^{-1}(\alpha)$ – standard normal quantile at confidence level $\alpha = 99\%$, i.e. $z_{0.99} = 2.326$;
- r_f – ECB deposit facility rate (annualised).

Remark 3.1. *For assets with fat-tailed return distributions, the Normal quantile may underestimate tail risk. A Cornish–Fisher expansion can be applied:*

$$z_\alpha^{CF} = z_\alpha + \frac{z_\alpha^2 - 1}{6}\,\hat{\gamma}_1 + \frac{z_\alpha^3 - 3z_\alpha}{24}\,\hat{\gamma}_2 - \frac{2z_\alpha^3 - 5z_\alpha}{36}\,\hat{\gamma}_1^2,$$

where $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are the estimated skewness and excess kurtosis of the return series.

Liquidity-Adjusted Haircut

An additional bid-ask spread adjustment is applied for assets with elevated transaction costs:

$$h_i^{\text{liq}} = h_i + \max\left(0,\ \frac{s_i - s^*}{2}\right), \tag{2}$$

where s_i is the observed time-weighted average bid-ask spread and $s^* = 0.10\%$ is the reference spread for liquid large-cap equities.

Indicative Haircut Table

Table **??** presents the operationally applied haircut schedule, derived from equation (**??**) and reviewed quarterly.

Table 1: Indicative Haircut Schedule by Asset Class

Asset Class	**Subcategory**	**Base Haircut h_i**
Government Bonds	AAA–AA, maturity < 1Y	0.5%
	AAA–AA, 1–5Y	2.0%
	AAA–AA, 5–10Y	4.0%
	A–BBB, any maturity	6–12%
IG Corporate Bonds	0–5Y	8.0%
	5Y+	12.0%
Large-Cap Equities	MSCI World, ADTV $> 5M$	20.0%
Mid-Cap Equities	ADTV $500K$–$5M$	30.0%
Small-Cap Equities	ADTV $< 500K$ (min)	40.0%
ETFs (broad)	Major index replication	20.0%
ETFs (thematic)	Sector/factor exposure	30.0%

3.3 Concentration Limits and Portfolio Adjustment

Single-Name Cap

The eligible collateral value attributed to any single issuer is capped at 20% of the unadjusted collateral value. Let $w_i = P_{i,t}Q_i(1 - h_i)/\sum_j P_{j,t}Q_j(1 - h_j)$ be the weight of issuer i. The concentration-adjusted contribution is:

$$C_i = P_{i,t}\,Q_i\,(1 - h_i) \cdot \min\left(1,\ \frac{0.20}{w_i}\right).$$

Portfolio Correlation Adjustment

Let $\bar{\rho}_t$ denote the average pairwise Pearson correlation of daily log-returns across all collateral positions, estimated over a 60-day rolling window. The correlation adjustment is:

$$\kappa(\bar{\rho}_t) = \begin{cases} 0 & \bar{\rho}_t \leq 0.4 \\ 0.05 \cdot \dfrac{\bar{\rho}_t - 0.4}{0.6} & 0.4 < \bar{\rho}_t \leq 1 \end{cases} \tag{3}$$

This adjustment linearly ramps from 0% to 5% as average pairwise correlation rises from 0.4 to 1.0, reflecting the reduced diversification benefit in stress.

4 LTV Calculation Methodology

4.1 Core LTV Formula

The adjusted collateral value incorporating all adjustments is:

$$V_t^{\text{adj}} = \big(1 - \kappa(\bar{\rho}_t)\big) \sum_{i=1}^{N} C_i(t), \tag{4}$$

and the LTV is:

$$\text{LTV}_t = \frac{L_t}{V_t^{\text{adj}}}. \tag{5}$$

4.2 Loan Balance Dynamics

The loan balance accrues continuously at the margin lending rate r_m (floating: EURIBOR 3M + contractual spread), reduced by any client repayments:

$$L_t = L_0\, e^{r_m t} - \sum_{k:\, t_k \leq t} R_k\, e^{r_m (t - t_k)}, \tag{6}$$

where R_k is the repayment made at time t_k.

4.3 Calculation Frequency and Price Sources

Table 2: LTV Calculation Schedule

Calculation	Frequency	Trigger
Real-time LTV	Continuous (tick)	Any price update
Intraday Snap	Every 15 min (market hrs)	Scheduled
End-of-Day LTV	Once, post close	Official close prices
Stress LTV	Daily, post close	Scheduled
Regime LTV	Daily, pre-open	Regime classification

Price source hierarchy:

1. Primary: Exchange last-traded price (real-time feed).

2. Fallback 1: Bloomberg/Refinitiv composite.
3. Fallback 2: Prior closing price with an additional 5% liquidity haircut.
4. If no valid price for > 2 consecutive trading days: zero collateral value.

5 Dynamic Threshold Adjustment

5.1 Motivation

Static LTV thresholds calibrated under average market conditions may be insufficient during periods of elevated volatility or structural regime change. A dynamic threshold framework adjusts the warning and liquidation LTVs in real time, ensuring that the probability of the loan becoming uncovered remains below the target tolerance p^* under all market regimes.

5.2 Volatility Regime Classification

Market regime is classified daily using a two-state Hidden Markov Model (HMM) estimated on the VSTOXX index (or CBOE VIX for US-dominated portfolios). Let the latent state $S_t \in \{0, 1\}$ denote *normal* and *stress* regimes respectively. The observation equation is:

$$x_t = \mu_{S_t} + \sigma_{S_t}\,\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1),$$

where $x_t = \ln(\text{VSTOXX}_t)$ and the parameters $\Theta = \{\mu_0, \sigma_0, \mu_1, \sigma_1, p_{00}, p_{11}\}$ are estimated via the Baum–Welch (EM) algorithm.

The smoothed state probability is:

$$\pi_t^{(1)} = \mathbb{P}(S_t = 1 \mid x_{1:t};\, \hat{\Theta}),$$

computed via the forward–backward algorithm.

In practice, a simpler but robust **three-regime classification** based on realised volatility percentiles is operationally preferred:

Table 3: Volatility Regime Classification

Regime	Condition	VSTOXX Level	Label
Normal	$\sigma_{p,5d} \le \bar{\sigma}_p + 1\hat{s}$	< 20	NORMAL
Elevated	$\bar{\sigma}_p + 1\hat{s} < \sigma_{p,5d} \le \bar{\sigma}_p + 2\hat{s}$	20–30	ELEVATED
Stress	$\sigma_{p,5d} > \bar{\sigma}_p + 2\hat{s}$	> 30	STRESS

where $\sigma_{p,5d}$ is the 5-day realised portfolio volatility, $\bar{\sigma}_p$ is the long-run mean, and $\hat{s}$ is the long-run standard deviation of 5-day realised volatility, both estimated over a 2-year rolling window.

5.3 Dynamic Threshold Formula

The regime-adjusted liquidation threshold is defined such that:

$$\mathbb{P}\Big(\Delta V_t > V_t^{\text{adj}}\big(1 - \text{LTV}_t^{\text{liq}}\big)\Big) \le p^*, \tag{7}$$

which, under a Normal approximation with portfolio volatility $\hat{\sigma}_{p,t}$ and liquidation horizon Δt, yields:

$$\mathrm{LTV}_t^{\mathrm{liq}} = 1 - z_{1-p^*}\,\hat{\sigma}_{p,t}\,\sqrt{\Delta t}, \tag{8}$$

where $\hat{\sigma}_{p,t}$ is updated daily. To avoid excessive threshold oscillation, the dynamic threshold is smoothed via an exponential moving average with half-life $\tau = 5$ trading days:

$$\widetilde{\mathrm{LTV}}_t^{\mathrm{liq}} = \beta\,\mathrm{LTV}_t^{\mathrm{liq}} + (1-\beta)\,\widetilde{\mathrm{LTV}}_{t-1}^{\mathrm{liq}}, \qquad \beta = 1 - e^{-\ln 2/\tau}. \tag{9}$$

Hard floors and ceilings are applied to ensure operational bounds:

$$\mathrm{LTV}_t^{\mathrm{liq},\star} = \min\Big(\mathrm{LTV}^{\mathrm{liq,max}},\ \max\Big(\mathrm{LTV}^{\mathrm{liq,min}},\ \widetilde{\mathrm{LTV}}_t^{\mathrm{liq}}\Big)\Big). \tag{10}$$

5.4 Regime-Dependent Threshold Schedule

Table **??** summarises the operationally applied thresholds across regimes, derived from equation (**??**) for a representative diversified equity portfolio ($p^* = 0.1\%$):

Table 4: Regime-Dependent LTV Threshold Schedule

Regime	$\hat{\sigma}_{p,t}$ (ann.)	**Initial LTV**	**Warning LTV**	**Liquidation LTV**
Normal	$\approx 15\%$	60%	70%	80%
Elevated	$\approx 25\%$	55%	65%	75%
Stress	$\approx 40\%$	45%	55%	65%

In the Stress regime, newly originated loans must satisfy the lower Initial LTV. Existing loans are not force-cured solely due to a regime transition, but receive an accelerated margin call cure deadline.

5.5 Asymmetric Regime Transitions

To avoid threshold whipsawing around regime boundaries, asymmetric persistence rules are applied:

- **Upgrade** (Stress $\rightarrow$ Elevated or Normal): requires 5 consecutive trading days in the lower-volatility regime before thresholds are relaxed.
- **Downgrade** (Normal/Elevated $\rightarrow$ Stress): immediate threshold tightening on the day of classification.

6 Margin Call Process

6.1 Trigger and Notification

A margin call is triggered when:

$$\mathrm{LTV}_t \geq \mathrm{LTV}_t^{\mathrm{warn},\star}.$$

The notification is issued within 15 minutes of trigger detection and specifies the current LTV, the warning threshold, the required cure amount Δ, and the cure deadline.

6.2 Cure Amount Calculation

The cure target is restoration to the regime-adjusted initial LTV, $\mathrm{LTV}^{\mathrm{target}} = \mathrm{LTV}_t^{\mathrm{init},\star}$. Three cure mechanisms are available:

Option A – Cash Deposit / Partial Repayment

$$\Delta^{\mathrm{cash}} = L_t - \mathrm{LTV}^{\mathrm{target}} \cdot V_t^{\mathrm{adj}}.$$

Option B – Additional Collateral

$$\Delta^{\mathrm{collateral}} = \frac{L_t - \mathrm{LTV}^{\mathrm{target}} \cdot V_t^{\mathrm{adj}}}{\mathrm{LTV}^{\mathrm{target}} \cdot (1 - h_{\mathrm{new}})},$$

where h_{new} is the haircut of the newly deposited collateral.

Option C – Partial Position Reduction

$$\Delta^{\mathrm{reduce}} = \frac{L_t - \mathrm{LTV}^{\mathrm{target}} \cdot V_t^{\mathrm{adj}}}{1 - \mathrm{LTV}^{\mathrm{target}} \cdot (1 - h_{\mathrm{sold}})},$$

where h_{sold} is the haircut of the sold collateral.

6.3 Cure Period

Client Type	Normal Regime	Stress Regime
Retail	T+1, 12:00 CET	Same-day, 4 hours
Professional	T+0, 4 hours	T+0, 2 hours

7 Forced Liquidation Process

7.1 Trigger Conditions

Forced liquidation is initiated when *any* of the following conditions hold:

1. $\mathrm{LTV}_t \geq \mathrm{LTV}_t^{\mathrm{liq},\star}$ at any intraday snapshot;
2. Cure period expires without sufficient cure action;
3. $\mathrm{LTV}_t^{\mathrm{stress}} \geq \mathrm{LTV}_t^{\mathrm{liq},\star}$ on two consecutive business days (pre-emptive trigger).

7.2 Liquidation Sequencing

Positions are ranked by liquidity score:

$$\ell_i = \frac{\mathrm{ADTV}_i}{P_{i,t}\, Q_i},$$

and liquidated in descending order of ℓ_i. The quantity of asset i to sell is:

$$Q_i^{\mathrm{sell}} = \min\left(Q_i,\ \frac{\Delta^{\mathrm{reduce}}}{P_{i,t}(1 - s_i/2)}\right),$$

where $s_i/2$ is the estimated half-spread. The algorithm iterates until $\mathrm{LTV}_t \leq \mathrm{LTV}^{\mathrm{target}}$.

7.3 Market Impact Estimation

For positions where $Q_i^{\text{sell}}/\text{ADTV}_i > 10\%$, market impact is estimated using the Almgren–Chriss permanent impact model:

$$MI_i = \eta\,\sigma_i\,\frac{Q_i^{\text{sell}}}{\text{ADTV}_i}\,P_{i,t}, \tag{11}$$

where $\eta \in [0.1, 0.3]$ is the market impact coefficient, estimated empirically. The total liquidation shortfall estimate is:

$$\text{TIS} = \sum_i MI_i + \sum_i P_{i,t}\,Q_i^{\text{sell}}\,\frac{s_i}{2}.$$

This shortfall is factored into the target cure amount.

7.4 Proceeds Waterfall

Liquidation proceeds are applied in strict priority:

1. Outstanding loan principal L_t;
2. Accrued interest;
3. Platform liquidation fee;
4. Residual returned to client.

8 Stress Testing Framework

8.1 Overview

The stress testing framework serves three distinct purposes:

1. **Pre-emptive monitoring:** Daily computation of Stress LTV to detect latent vulnerabilities before mark-to-market LTV breaches thresholds.
2. **Threshold calibration:** Stress scenarios inform the setting of regime-adjusted thresholds (Section **??**).
3. **Lender capital planning:** Portfolio-level stress results are reported to the Lender for concentration risk and capital adequacy assessment.

8.2 Single-Asset Stress LTV

For each client portfolio, a Stress LTV is computed by applying instantaneous shocks δ_i to all collateral positions:

$$V_t^{\text{stress}} = \sum_{i=1}^{N} P_{i,t}\,Q_i\,(1-h_i)\,(1-\delta_i), \qquad \text{LTV}_t^{\text{stress}} = \frac{L_t}{V_t^{\text{stress}}}. \tag{12}$$

The stress shock δ_i is defined as:

$$\delta_i = \max\bigl(\delta_i^{\text{hist}},\ \delta_i^{\text{floor}}\bigr)\,,$$

where:

- δ_i^{hist} is the 99th percentile of the empirical distribution of 5-day absolute log-returns of asset i over a 5-year lookback:

$$\delta_i^{\text{hist}} = \widehat{F}_i^{-1}(0.99), \qquad \widehat{F}_i = \text{empirical CDF of } |r_{i,t:t+5\Delta}|;$$

- $\delta_i^{\text{floor}} = 15\%$ for equities, 5% for investment-grade bonds (regulatory floor).

8.3 Historical Scenario Analysis

The collateral portfolio is revalued under the following named historical stress scenarios, replaying actual observed returns:

Table 5: Named Historical Stress Scenarios

Scenario	Period	Approx. Peak Equity Drawdown
Global Financial Crisis	Oct 2008 – Mar 2009	−50%
European Sovereign Crisis	Aug 2011 – Sep 2011	−22%
COVID-19 Shock	Feb 2020 – Mar 2020	−35%
Rate Shock 2022	Jan 2022 – Oct 2022	−25% (equities), −20% (bonds)
Flash Crash	Aug 24, 2015	−11% intraday

For each scenario s with path $\{r_{i,t}^{(s)}\}$, the portfolio trajectory is:

$$V_\tau^{(s)} = \sum_{i=1}^{N} P_{i,0}\, Q_i\, (1-h_i) \prod_{t=1}^{\tau} \big(1 + r_{i,t}^{(s)}\big), \qquad \text{LTV}_\tau^{(s)} = \frac{L_\tau}{V_\tau^{(s)}},$$

and the scenario breach indicator is:

$$\mathbb{1}_{\text{breach}}^{(s)} = \max_{\tau} \mathbb{1}\Big[\text{LTV}_\tau^{(s)} \geq \text{LTV}^{\text{liq},\star}\Big].$$

8.4 Hypothetical (Adverse) Scenario Analysis

In addition to historical replays, the following hypothetical shocks are applied:

1. **Parallel equity drawdown:** All equity positions simultaneously lose $X\%$ over one trading day. X is swept from 10% to 50%.
2. **Rates shock:** Instantaneous parallel shift of +200bps applied to bond positions (duration-adjusted price impact).
3. **Correlation breakdown:** All pairwise correlations set to 1.0; haircuts are recomputed accordingly and $\kappa = 5\%$ applied.
4. **Liquidity seizure:** All ADTV values reduced by 80%; haircuts recalculated via equation (**??**) with inflated spreads.
5. **FX shock:** Non-EUR assets subject to an additional 15% adverse currency move.

8.5 Reverse Stress Testing

Reverse stress testing identifies the minimum market shock δ^* required to breach the liquidation threshold, i.e. to make the loan uncovered. This is formulated as a constrained optimisation:

$$\delta^* = \arg\min_{\boldsymbol{\delta} \geq 0} \|\boldsymbol{\delta}\|_2 \quad \text{s.t.} \quad L_t \geq V_t^{\text{adj}}(\boldsymbol{\delta}), \tag{13}$$

where $\|\boldsymbol{\delta}\|_2 = \sqrt{\sum_i \delta_i^2}$ is the Euclidean norm of the shock vector and:

$$V_t^{\text{adj}}(\boldsymbol{\delta}) = \big(1-\kappa\big) \sum_{i=1}^{N} P_{i,t}\, Q_i\, (1-h_i)\, (1-\delta_i).$$

The closed-form solution, assuming no binding concentration constraints, is:

$$\delta_i^* = \delta_{\text{uniform}}^* = 1 - \frac{V_t^{\text{adj}}(0)}{L_t/(1-\kappa)} \cdot \frac{1}{N}, \tag{14}$$

for a uniform shock. For a risk-weighted shock (proportional to volatility):

$$\delta_i^* = \lambda^*\, \sigma_i, \qquad \lambda^* = \frac{V_t^{\text{adj}}(0) - L_t/(1-\kappa)}{\sum_j P_{j,t}\, Q_j\, (1-h_j)\, \sigma_j}.$$

The reverse stress result δ^* is reported alongside each daily Stress LTV as a margin of safety metric.

8.6 Portfolio-Level VaR and CVaR of LTV

For the purpose of Lender reporting, a distribution of $\text{LTV}_{t+\Delta t}$ is estimated via Monte Carlo simulation:

1. Draw $M = 10{,}000$ correlated return scenarios from a multivariate t-distribution with estimated covariance $\hat{\Sigma}$ and degrees of freedom ν (fitted via maximum likelihood):

$$\mathbf{r}_{t+\Delta t}^{(m)} \sim t_\nu(\mathbf{0},\, \hat{\Sigma}\,\Delta t), \qquad m = 1, \ldots, M.$$

2. Revalue the portfolio under each scenario:

$$V_{t+\Delta t}^{(m)} = \sum_{i=1}^{N} P_{i,t}\, e^{r_{i,t+\Delta t}^{(m)}}\, Q_i\, (1-h_i).$$

3. Compute the simulated LTV:

$$\text{LTV}_{t+\Delta t}^{(m)} = \frac{L_{t+\Delta t}}{V_{t+\Delta t}^{(m)}}.$$

4. Report:

$$\text{VaR}_\alpha(\text{LTV}) = \hat{F}_{\text{LTV}}^{-1}(\alpha),$$
$$\text{CVaR}_\alpha(\text{LTV}) = \frac{1}{1-\alpha} \int_\alpha^1 \hat{F}_{\text{LTV}}^{-1}(u)\, du.$$

9 Collateral Revaluation and Corporate Actions

Table 6: Corporate Action Treatment

Event	**Treatment**
Cash Dividend	Ex-date price drop reflected via feed; dividend cash credited and may offset loan
Stock Split	$Q_i \to kQ_i$, $P_i \to P_i/k$; LTV unchanged if processed correctly
Spin-off	New entity assessed for eligibility; if ineligible, zero collateral value from ex-date
M&A (cash)	Position removed on settlement; proceeds applied to loan
M&A (stock)	New shares reassessed; LTV recalculated
Delisting	Immediate zero collateral value; margin call if $\mathrm{LTV} \geq \mathrm{LTV}^{\mathrm{warn}}$
Trading Halt	Fallback price applied; zero value if halt > 2 days

9.1 FX Revaluation

For assets denominated in currency $c \neq$ EUR, a combined haircut is applied:

$$h_i^{\mathrm{total}} = 1 - (1 - h_i)(1 - h_{\mathrm{FX}(c)}),$$

where $h_{\mathrm{FX}(c)}$ is derived from equation (**??**) applied to the c/EUR exchange rate with a 5-day liquidation horizon.

10 Governance and Validation

10.1 Backtesting and the Kupiec POF Test

Haircut adequacy is validated quarterly. For each asset class j, the test statistic is:

$$LR_{\mathrm{POF}}^{(j)} = -2\ln\left[\frac{(1-p)^{T-x}\,p^x}{(1-\hat{p})^{T-x}\,\hat{p}^x}\right] \xrightarrow{d} \chi^2(1), \tag{15}$$

where p is the target exceedance rate ($p = 1\%$ for $\alpha = 99\%$), T is the sample size, x is the number of observed haircut breaches, and $\hat{p} = x/T$. Haircuts are flagged for revision if $LR > \chi^2_{0.95}(1) = 3.84$.

Additionally, the Christoffersen conditional coverage test is applied to assess independence of exceedances:

$$LR_{\mathrm{CC}} = LR_{\mathrm{POF}} + LR_{\mathrm{ind}} \;\sim\; \chi^2(2).$$

10.2 Review Schedule

Activity	Frequency
Haircut backtesting (Kupiec + Christoffersen)	Quarterly
Regime model recalibration (HMM)	Semi-annual
Threshold calibration review	Quarterly
Stress scenario update	Annual (or post-crisis)
Full methodology review	Annual
Lender portfolio LTV report	Daily
Lender stress report	Weekly

A Notation Summary

Symbol	Description
L_t	Loan amount at time t
V_t^{gross}	Gross market value of collateral at t
V_t^{adj}	Adjusted collateral value at t
V_t^{stress}	Stressed collateral value at t
$P_{i,t}$	Price of asset i at t
Q_i	Quantity held of asset i
h_i	Haircut for asset i
h_i^{liq}	Liquidity-adjusted haircut for asset i
$h_{\text{FX}(c)}$	FX haircut for currency c
$\hat{\sigma}_i$	EWMA annualised volatility of asset i
$\hat{\sigma}_{p,t}$	Portfolio volatility at t
$\bar{\rho}_t$	Average pairwise correlation at t
$\kappa(\bar{\rho}_t)$	Portfolio correlation adjustment
S_t	Volatility regime state at t
$\pi_t^{(1)}$	Smoothed stress-regime probability
δ_i	Stress shock for asset i
δ^*	Reverse stress break-even shock
ℓ_i	Liquidity score of asset i
r_m	Margin loan rate
r_f	Risk-free rate
Δt	Liquidation horizon (years)
z_α	Standard normal quantile at level α
λ	EWMA decay factor
η	Almgren–Chriss impact coefficient
p^*	Target loss probability tolerance
LTV^{init}	Maximum initial LTV
LTV^{warn}	Warning LTV threshold
LTV^{liq}	Liquidation LTV threshold
$\text{LTV}_t^{\text{liq},\star}$	Dynamic (regime-adjusted) liquidation threshold

B Worked Example: Dynamic Threshold and Stress LTV

Portfolio

- 1,000 shares large-cap equity: $P = 80$, $h = 20\%$, $\sigma = 18\%$ p.a.
- 20,000 nominal 3Y AA government bond: price 101%, $h = 2\%$, $\sigma = 3\%$ p.a.
- 500 shares mid-cap equity: $P = 40$, $h = 30\%$, $\sigma = 28\%$ p.a.

$$V^{\text{adj}} = 64{,}000 + 19{,}796 + 14{,}000 = 97{,}796, \qquad L = 55{,}000, \qquad \text{LTV} = 56.2\%.$$

Portfolio Volatility

Assuming correlations $\rho_{12} = 0.05$, $\rho_{13} = 0.60$, $\rho_{23} = 0.05$, portfolio weights (by adjusted value) $w_1 = 65.4\%$, $w_2 = 20.2\%$, $w_3 = 14.3\%$:

$$\hat{\sigma}_p^2 = \mathbf{w}^\top \Sigma \mathbf{w} \approx 0.0256 \;\Rightarrow\; \hat{\sigma}_p \approx 16.0\% \text{ p.a.}$$

Dynamic Liquidation Threshold (Normal Regime)

$$\text{LTV}^{\text{liq}} = 1 - 3.09 \times 0.160 \times \sqrt{5/252} = 1 - 0.1754 = 82.5\%.$$

After smoothing and applying bounds: $\text{LTV}^{\text{liq},\star} = 80\%$.

Stress LTV

$\delta_1^{\text{hist}} = 22\%$, $\delta_2^{\text{hist}} = 5\%$, $\delta_3^{\text{hist}} = 28\%$:

$V^{\text{stress}} = 64{,}000\times(1-0.22)+19{,}796\times(1-0.05)+14{,}000\times(1-0.28) = 49{,}920+18{,}806+10{,}080 = 78{,}806.$

$$\text{LTV}^{\text{stress}} = \frac{55{,}000}{78{,}806} = \mathbf{69.8\%}.$$

The Stress LTV is below $\text{LTV}^{\text{liq},\star} = 80\%$: **no pre-emptive trigger**.

Reverse Stress Break-Even

The uniform shock required to make $L \geq V^{\text{adj}}$:

$$\delta^* = 1 - \frac{L}{V^{\text{adj}}} = 1 - \frac{55{,}000}{97{,}796} = 43.8\%.$$

A simultaneous 43.8% uniform drawdown across all positions would render the loan uncovered. This exceeds the GFC peak drawdown of $\approx 35\%$, indicating adequate stress resilience for this portfolio.